JAYAGNA HIREN PATEL

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Education

Birla Institute of Technology and Science, Pilani Campus (BITS Pilani) **2023 – May 2028 (expected)**
B.E. Computer Science Engineering + M.Sc. Mathematics **CGPA: 8.28/10**

Relevant Coursework: Data Structures and Algorithms, Probability and Statistics, Deep Learning, Database Management Systems, Optimization, Operations Research, Graphs and Networks

Technical Skills

Programming: Python, TypeScript, Java, SQL, C
Core: Object-Oriented Programming (OOP), Data Structures and Algorithms, Optimization, Graph Algorithms
AI/ML: Scikit-learn, TensorFlow, XGBoost, YOLO11, NumPy, Machine Learning, Deep Learning, Computer Vision
Generative AI & Retrieval: Gemini, LangChain, RAG, pgvector, Pinecone, Embeddings, Vector Search
Backend & Cloud: FastAPI, Next.js, Supabase, PostgreSQL, Spring Boot, REST APIs, AWS, Redis, Git, CI/CD

Internships

GPLAN **Jul 2025 – Jan 2026**
Machine Learning Intern : Optimization Algorithms
- Developed a **genetic algorithm** to optimize room placement under geometric, spacing, and layout constraints.
- Designed **fitness, selection, crossover, and mutation** logic for iterative floorplan optimization.
- Implemented **geometry and graph algorithms** for overlap detection, validity checks, and room connectivity.

Amukha Technologies Private Limited **May 2025 – Jul 2025**
Credit Risk Analyst Intern
- Built a **Loan Pricing Index** using **ML and financial analytics** to forecast lending rates from borrower data.
- Performed **feature engineering, regression modeling, and validation** across structured real loan datasets.
- Deployed a local ML application for **real time interest prediction** from user provided lending inputs.

Projects

Nishtha Physiotherapy: Clinic Website + PhysioBOT | Live | Next.js, Supabase, FastAPI **Sep 2024 – Oct 2026**
- Built and deployed the **booking site and admin panel** for a clinic seeing **∼150 patients/day**.
- Productionized my RAG prototype as **PhysioBOT**: page-cited answers in **English, Gujarati and Hindi**.
- **Rule-based emergency triage** bypasses the LLM; prompt-injection guards, per-IP rate limits, **31 tests** in CI.
- Protected patient data with **row-level security**; a **partial unique index** blocks double bookings.

SmogCast: 72-hour PM2.5 Forecasting for Delhi NCR | Python, scikit-learn, Streamlit **Oct 2026**
- Built a **multi-horizon gradient-boosting** forecaster on **OpenAQ, ERA5 and NASA FIRMS** data (**15 stations**).
- On a held-out year, cut next-day PM2.5 error **19%** vs persistence and **42%** vs climatology (**65%** AQI accuracy).
- Estimated **crop-fire smoke** via counterfactuals (up to **16%** of PM2.5); daily **GitHub Actions** job + dashboard.

Career Mode Scout: FIFA Transfer Planning | Python, scikit-learn, HiGHS, Streamlit **Sep 2026 – Oct 2026**
- Beat EA's potential rating by **15%** on 2-year forecasts using **gradient boosting** on **139k FIFA player-seasons**.
- Built an exact **MILP squad optimizer** (HiGHS) that beat a greedy strategy in **60%** of **120** transfer scenarios.
- Built a **similarity-search replacement finder**: same-position players **63% cheaper** for **89%** of 80+ stars.

VigiLens (Traffic Violation Detection System) **Apr 2026 – May 2026**
- Built a **YOLO11 computer vision pipeline** to detect vehicles, pedestrians, and **3 traffic violation categories**.
- Applied **spatial rule analysis** for stopline, overcrowding, and wrong side driving detection.
- Designed a **cascading OCR pipeline** with **4 preprocessing variants** for license plate recognition and validation.

Leadership & Activities

AI/ML Lead – ACM BITS Pilani Student Chapter **Jul 2025 – Jul 2026**
- Led AI/ML initiatives for **50+ students** through technical workshops and machine learning project guidance.
- Organized **10+ AI/ML learning sessions**, curated technical resources, and collaborated with faculty to support **5+ student projects** from ideation to implementation using modern ML tools.

BITS Pilani Hockey Team **Bronze Medal – BOSM 2025**
Represented BITS Pilani competitively and contributed to **team strategy and match performance**.